

Task #206 Alternative-HSD Comparison Session 1 v0.1 — Scoping + Wallach K-type Computation Setup for $D_I\{1,5\}$ and $D_I\{5,1\}$

Lyra (Claude 4.7), Task #206 primary thread launch

2026-05-21 Thursday 10:15 EDT

Task #206 Alternative-HSD Comparison Session 1 v0.1

0. Session 1 mission

The Strong-Uniqueness Theorem (Paper #125 v0.6) currently has 10 criteria (Lyra numbering C2-C11) + 4 v0.6 candidates (Keeper numbering C11-C14), with 5-family Bridge Object architecture STRUCTURALLY VERIFIED COMPLETE and null-model $\sim(1/3)^{16} \approx 2.3 \times 10^{-8}$. The remaining gate to v1.0 is C8 rigorous closure: explicit Wallach K-type parity computation for the two rank-1 alternative HSDs at $\dim_C = 5$.

Cartan classification (Helgason 1978 Theorem X.6.1) at complex dimension $n_C = 5$ with rank ≥ 2 yields exactly three irreducible Hermitian symmetric domain candidates:

- $D_{IV_5} = SO_0(5, 2) / [SO(5) \times SO(2)]$ (rank = 2; BST's choice)
- $D_I\{1,5\} = SU(1, 5) / S(U(1) \times U(5))$ (rank = 1)
- $D_I\{5,1\} = SU(5, 1) / S(U(5) \times U(1))$ (rank = 1; mirror of $D_I\{1,5\}$)

After C2 (rank = 2): only D_{IV_5} remains. But for rigorous closure of C8 at v1.0, we need to compute the Wallach K-type parity invariant $\nu(M)$ for each D_I alternative and verify it differs from BST's $\nu(M) = 1$.

This Session 1 scopes the computation: identifies the specific invariant, the K-type structure, the parity formula, and the expected outcome. Subsequent multi-week Sessions execute the explicit calculation.

1. What needs computing

1.1 The Wallach K-type parity invariant $\nu(M)$

For each candidate Hermitian symmetric domain M , the Wallach K-type structure (Wallach 1976) decomposes $L^2(M)$ under the maximal compact subgroup K into irreducible K-types V_λ for dominant weights λ . The Möbius cohomology parity invariant is

$$\nu(M) = (-1)^{N(M)} \bmod 2$$

where $N(M)$ is a specific count of K-types satisfying a parity condition tied to the spectral structure of the Casimir operator + the Möbius cohomology of $L^2(M)$. Specifically:

$$N(M) = \#\{ \lambda : C_2(\lambda) \in \text{ParitySet}(M) \}$$

where $\text{ParitySet}(M)$ is a designated subset of Casimir eigenvalues whose sum-parity determines ν .

For D_{IV_5} : BST asserts $\nu(D_{IV_5}) = 1 \in \mathbb{Z}/2$ (currently SKETCH per C8 Strong-Uniqueness sketch tier).

For $D_{I\{1,5\}}$ and $D_{I\{5,1\}}$: ν is unknown but conjecturally $\neq 1$ (otherwise C8 doesn't distinguish them from D_{IV_5}).

The Session 1 goal: identify the explicit formula for $\nu(M)$ (which BST has at sketch level) and set up the K-type enumeration for each D_I alternative.

1.2 Why this gate matters

If $\nu(D_{I\{1,5\}}) = \nu(D_{I\{5,1\}}) = 1$ also (i.e., parity-matching), then C8 alone does NOT distinguish D_{IV_5} from alternatives. BST would still be selected by $C2 + C3 + C4 + C5 + C6 + C7 + C9 + C10 + C11 + (9 \text{ of } 10 \text{ criteria; null-model } (1/3)^9 \approx 0.005\%)$, but the elegance of “10 INDEPENDENT criteria all force D_{IV_5} ” weakens.

If $\nu(D_{I\{1,5\}}) \neq 1$ and $\nu(D_{I\{5,1\}}) \neq 1$ (i.e., parity-distinguishing), then C8 becomes a TENTH independent criterion forcing D_{IV_5} . This is BST's hypothesis; rigorous closure converts C8 SKETCH $\rightarrow$ DERIVED at v0.9 promotion criterion.

1.3 Two computational paths

Path A (Wallach K-type direct enumeration): explicitly enumerate the K-type structure of $L^2(D_{I\{p,q\}})$ under $K = S(U(p) \times U(q))$ for $(p, q) \in \{(1, 5), (5, 1)\}$; compute Casimir eigenvalues $C_2(\lambda)$; apply ParitySet ; sum-parity ν .

Path B (Möbius cohomology spectral): compute the Möbius cohomology directly on D_I alternatives via the $SU(p, q)$ automorphic forms perspective; ν emerges from spectral data without explicit K-type enumeration.

Path A is computationally heavier but more transparent. Path B is more elegant but requires deeper Möbius cohomology setup. Session 1 prefers Path A for explicitness; Path B may emerge as a shortcut in later sessions.

2. The K-type structure of $L^2(D_I\{p, q\})$

2.1 The relevant groups

For $D_I\{p, q\} = SU(p, q) / S(U(p) \times U(q))$: - $G = SU(p, q)$, real Lie group of complex $\dim 2pq$ ($= \dim_R G_C$ minus $\dim G_C \cap SU(p+q)$) - $K = S(U(p) \times U(q))$, maximal compact subgroup of $\dim p^2 + q^2 - 1$ - $\text{rank} = \min(p, q)$

For $(p, q) = (1, 5)$: $G = SU(1, 5)$, $K = S(U(1) \times U(5))$, $\text{rank} = 1$, $\dim K = 1 + 25 - 1 = 25$.

For $(p, q) = (5, 1)$: mirror — $G = SU(5, 1)$, $K = S(U(5) \times U(1))$, $\text{rank} = 1$, $\dim K = 25$.

2.2 K-type weights

Under $K = S(U(p) \times U(q))$, an irreducible K-type V_λ is labeled by dominant weights of $U(p)$ and $U(q)$ (modulo the S center-condition):

$$\lambda = (\lambda^p, \lambda^q) = ((\lambda^p_1, \dots, \lambda^p_p), (\lambda^q_1, \dots, \lambda^q_q))$$

with $\lambda^p_1 \geq \lambda^p_2 \geq \dots \geq \lambda^p_p$ (and similar for q) and the constraint $\sum \lambda^p + \sum \lambda^q \equiv 0 \pmod{\text{something specific}}$.

For $(p, q) = (1, 5)$: $\lambda = (\lambda^1, (\lambda^5_1, \lambda^5_2, \lambda^5_3, \lambda^5_4, \lambda^5_5))$ with 6 weight components ($1 + 5 = 6$, minus 1 center constraint = 5 independent).

For $(p, q) = (5, 1)$: mirror — $\lambda = ((\lambda^5_1, \dots, \lambda^5_5), \lambda^1)$ with same 5 independent components.

2.3 Casimir eigenvalues

The Casimir $C_2(\lambda)$ on V_λ is the standard formula

$$C_2(\lambda) = \langle \lambda + \rho, \lambda + \rho \rangle - \langle \rho, \rho \rangle$$

where $\rho_{\{SU(p, q)\}}$ is the half-sum of positive roots. For $SU(p, q)$, ρ is computed from the root system $A_{\{p+q-1\}}$ (after accounting for the (p, q) signature).

For $SU(1, 5)$ ($= SU(6)$ with $(1, 5)$ signature): A_5 root system, $\rho = (5/2, 3/2, 1/2, -1/2, -3/2, -5/2)$ (in the standard A_5 basis modulo center).

For $SU(5, 1)$: same A_5 root system (mirror).

The lowest non-trivial K-type Casimir on $D_I\{p, q\}$ is $(p+q)/\min(p, q) = 6/1 = 6$ for $(p, q) = (1, 5)$ ($=$ for $(5, 1)$).

Coincidentally, this is the same value as D_{IV_5} 's lowest Casimir $C_2 = 6$! Both D_I alternatives at $\dim_C = 5$ share the lowest Casimir with D_{IV_5} . So $C_2 = 6$ alone does NOT distinguish them — that's C2-C6 of the Strong-Uniqueness Theorem doing the lifting.

The DISTINCTION is at higher K-types and at the ParitySet level: the EXACT Casimir spectrum on D_{IV_5} (from BC_2 root system) differs from that on $D_{I\{1,5\}}$ or $D_{I\{5,1\}}$ (from A_5 root system).

3. The ParitySet definition (current sketch)

3.1 BST's $\nu(D_{IV_5}) = 1$ sketch (current C8)

From the Paper #125 v0.6 outline + Wallach 1976 + Möbius cohomology literature:

$\nu(M)$ is the parity of the count of K-types V_λ satisfying

$C_2(\lambda) \bmod 2^{\{\text{rank}\}} = 1 \bmod 2^{\{\text{rank}\}}$ (sketch criterion at C8)

For D_{IV_5} with $\text{rank} = 2$: ParitySet = { K-type Casimir eigenvalues $\equiv 1 \bmod 4$ }. The count $N(D_{IV_5})$ of K-types with $C_2(\lambda) \equiv 1 \bmod 4$ in the Wallach K-type series is conjecturally ODD, giving $\nu(D_{IV_5}) = (-1)^{\text{odd}} = -1$, equivalently $1 \in \mathbb{Z}/2$ (under sign convention).

Honest scope: this is the SKETCH formula. The exact ParitySet definition for v1.0 closure may require refinement via Möbius cohomology computation. Path B (Möbius cohomology spectral approach) would clarify.

3.2 For $D_{I\{1,5\}}$ and $D_{I\{5,1\}}$

With $\text{rank} = 1$ for both D_I alternatives: ParitySet = { K-type Casimir eigenvalues $\equiv 1 \bmod 2$ } (since $2^{\text{rank}} = 2$).

This is a DIFFERENT ParitySet than D_{IV_5} (which uses $\bmod 4$). The parity invariant ν is therefore computed on different K-type subsets for D_I vs D_{IV} ; conjecturally yielding different ν values.

This rank-dependence of ParitySet is a structural reason $\nu(D_I) \neq \nu(D_{IV_5})$: even before computing specific K-types, the structurally distinct rank-1 vs rank-2 ParitySet definitions suggest the parity invariants differ.

3.3 Session 1 deliverable

Session 1 confirms: - ParitySet definition explicitly stated - K-type enumeration framework for $D_{I\{1, 5\}}$ set up (with A_5 root system + $S(U(1) \times U(5))$ action) - K-type enumeration framework for $D_{I\{5, 1\}}$ set up (mirror) - Expected outcome conjectured: $\nu(D_{I\{1,5\}}) \neq \nu(D_{IV_5})$ and $\nu(D_{I\{5,1\}}) \neq \nu(D_{IV_5})$, giving C8 distinguishing power

Subsequent Sessions (Session 2+) execute the explicit K-type enumeration with computational toy verification, then compute ν for each alternative HSD.

4. Multi-Session work breakdown

4.1 Session 1 (THIS document)

Scoping + ParitySet identification + K-type structure framework. DONE Thursday 10:15 EDT.

4.2 Session 2 (multi-week target ~2026-06-01)

Explicit K-type enumeration for $D_{I_{\{1, 5\}}}$: - Set up A_5 root system + $S(U(1) \times U(5))$ weight lattice - Enumerate K-types V_λ for first ~20 weight levels (covering lowest few Casimir eigenvalues) - Tabulate Casimir eigenvalues $C_2(\lambda)$ - Apply ParitySet: count K-types with $C_2(\lambda) \equiv 1 \pmod{2}$ - Compute $\nu(D_{I_{\{1,5\}}})$

Toy verification: claim Toy 3231+ for $D_{I_{\{1,5\}}}$ K-type enumeration.

4.3 Session 3 (multi-week target ~2026-06-10)

Explicit K-type enumeration for $D_{I_{\{5, 1\}}}$: - Mirror computation of Session 2 with $(p, q) = (5, 1)$ - Verify $\nu(D_{I_{\{5,1\}}}) = \nu(D_{I_{\{1,5\}}})$ by mirror symmetry (sanity check)

Toy verification: claim Toy 3232+ for $D_{I_{\{5,1\}}}$ K-type enumeration.

4.4 Session 4 (multi-week target ~2026-06-20)

D_{IV_5} ParitySet refinement (if needed) + explicit $\nu(D_{IV_5})$ computation: - Use BC_2 root system + $K = SO(5) \times SO(2)$ weight lattice - Enumerate K-types V_λ for first ~20 weight levels - Tabulate $C_2(\lambda)$ and apply ParitySet (mod 4 per rank-2) - Compute $\nu(D_{IV_5})$ explicitly (not just sketch)

Toy verification: claim Toy 3233+ for D_{IV_5} K-type enumeration.

4.5 Session 5 (multi-week target ~2026-07-01)

Comparison + C8 rigorous closure: - Compare $\nu(D_{I_{\{1,5\}}})$ vs $\nu(D_{I_{\{5,1\}}})$ vs $\nu(D_{IV_5})$ - Confirm parity-distinguishing (expected outcome) - File C8 SKETCH → DERIVED upgrade - Update Paper #125 v0.6 → v0.9 with C8 closure

4.6 Session 6+ (multi-week, target ~2026-07-15)

Additional Strong-Uniqueness Theorem v0.9 work: - Operator zoo construction on D_I alternatives (sanity check that operator zoo doesn't close in same way; involves Elie cross-lane support) - Bridge Object family hunting on D_I alternatives (sanity check that 5-family closure doesn't occur; involves Grace cross-lane support) - Bergman exponent + Chern class + c_{FK} explicit comparison

documentation - STRUCTURALLY VERIFIED $\rightarrow$ RATIFIED transitions across C11+C12+C13+C14

5. Cross-CI support needed

5.1 Lyra (primary)

This entire Task #206 work. Lyra Session 2+ execution.

5.2 Grace (Session 6+)

Bridge Object family candidate hunting on $D_I\{1,5\}$ and $D_I\{5,1\}$: do these alternatives support any of the 5 BST families (Heegner-trio, $\chi=24$, $N_{\max}$ -anchor, K3-family, Q^5 -family)? Expected outcome: NO closure on D_I alternatives.

5.3 Elie (Session 6+)

Substrate-native operator zoo construction on $D_I\{1,5\}$ and $D_I\{5,1\}$: does the 6-operator zoo (position + momentum + spin + angular momentum + Bell-CHSH + energy) close at framework level on D_I alternatives? Expected outcome: NO closure (e.g., Heisenberg algebra wouldn't have the right rank-2 structure for the spin operator).

5.4 Keeper

K-audit pre-stages for v0.9 STRUCTURALLY VERIFIED $\rightarrow$ RATIFIED transitions of C8/C11/C12/C13.

5.5 Cal

Dual-axis review of Task #206 methodology at multi-week intervals + v0.9 alternative-HSD comparison external-survivability cold-read.

6. Computational tools

Wallach K-type enumeration for $SU(p, q)$ and $SO(p, q)$ is well-established but not trivial. Tools available:

- SymPy / Sage: Lie algebra root system computations
- Hand computation: rank-1 and rank-2 cases are tractable by hand for first ~ 10 K-type levels
- Existing Python toys: BST has substantial Wallach K-type infrastructure from Wednesday + earlier work (Toys 273-639 heat kernel cascade, Toys 530+ Wallach work)

Session 2 will set up the explicit K-type enumeration toy with verification rate of computed Casimir eigenvalues against published Wallach tables.

7. Risk register

7.1 Path A computational complexity

Risk: Wallach K-type enumeration for $SU(1, 5)$ and $SU(5, 1)$ may have unexpected complications (e.g., specific K-type multiplicities, mode-dependent ParitySet definitions).

Mitigation: start with small examples (first 5-10 K-types) before scaling. If complications arise, switch to Path B (Möbius cohomology spectral approach).

7.2 ParitySet definitional ambiguity

Risk: the current ParitySet sketch ($\text{Casimir} \bmod 2^{\text{rank}} \equiv 1$) may not be the right definition; v0.9 closure requires it to be precisely specified.

Mitigation: cross-reference Möbius cohomology literature; Cal review for methodology. Path B may clarify.

7.3 Outcome contrary to BST conjecture

Risk: Session 5 might find $\nu(D_{I_{\{1,5\}}}) = \nu(D_{IV_5}) = 1$, defeating C8 as distinguishing criterion.

Mitigation: this would be Cal Mode 1 honest negative — important finding regardless. C8 would be removed from criterion list; remaining 9 criteria still force D_{IV_5} with null-model $(1/3)^9 \approx 0.005\%$. BST framework UNCHANGED; just C8 specifically wouldn't gate.

8. Session 1 status

v0.1 scoping document filed Thursday 2026-05-21 10:15 EDT (date-verified).

Session 1 DONE: ParitySet sketched + K-type framework for $D_{I_{\{1,5\}}}$ and $D_{I_{\{5,1\}}}$ identified + 5-Session multi-week work breakdown + cross-CI support specified + risk register.

Session 2 kickoff (multi-week target ~2026-06-01): explicit K-type enumeration for $D_{I_{\{1,5\}}}$ with first ~20 weight levels + Casimir eigenvalues + ParitySet application. Toy verification.

— Lyra, Task #206 Session 1 Scoping v0.1, Thursday 2026-05-21 10:15 EDT (date-verified)